Representative Publications

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2026 **SVoT: State-aware Visualization-of-Thought for Spatial Reasoning via Reinforcement Learning**

C. Lei, Y. Jiang, M. Hiller, Z. Zhou, X. Tian, K. A. Ehinger, N. Lipovetzky

The Fortieth Annual Conference on Neural Information Processing Systems, 2026 (NeurIPS).

2026 **From Reasoning to Generalization: Knowledge-Augmented LLMs for ARC Benchmark**

C. Lei, K. A. Ehinger, N. Lipovetzky, Y. Chang.

Conference on Empirical Methods in Natural Language Processing, 2026 (EMNLP Main Track).

2026 **Mind the Perspective: Let's Reason Recursively for Theory of Mind**

C. Lei, G. Hu, M. Yang, Y. Jiang, N. Lipovetzky.

Findings of the Conference on Empirical Methods in Natural Language Processing, 2026 (EMNLP Findings).

2026 **CARE: Grounded Recursive Self-Improvement for Production Customer-Service Agents**

Y. Jiang, **C. Lei**, R. Gao, X. Zhao, Y. Ge, H. Liu, R. Nie, et al. *arXiv preprint, 2026*

2025 **Planning-Driven Programming: A Large Language Model Programming Workflow**

C. Lei, Y. Chang, N. Lipovetzky, K. A. Ehinger.

Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025 (ACL Main Track).

2025 **State-Based Disassembly Planning**

C. Lei, N. Lipovetzky, K. A. Ehinger.

Proceedings of the AAAI Conference on Artificial Intelligence, 2025 (AAAI).

2024 **Generalized Planning for the Abstraction and Reasoning Corpus**

C. Lei, N. Lipovetzky, K. A. Ehinger.

Proceedings of the AAAI Conference on Artificial Intelligence, 2024 (AAAI).